

Co-UPP: A Co-optimized UAV Path Planning Framework for Balancing Truth Discovery and Environmental Sensing in Mobile CrowdSensing

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Abstract—Mobile CrowdSensing (MCS) systems are often hampered by ineffective task allocation and inefficient resource utilization, which largely stems from a lack of real-time environmental awareness. Although Unmanned Aerial Vehicles (UAV) present promising opportunities, their application is often limited to sparse ground truth collection, leaving their broader sensing capabilities underutilized. To address these limitations, we propose a Co-Optimized UAV Path Planning (Co-UPP) framework that improves MCS effectiveness through dynamic environmental sensing. The Co-UPP framework makes two primary contributions. First, it introduces a multi-objective path planning algorithm, termed the Evolutionary Fusion Dual-stage Hybrid (EF-DH) algorithm. The EF-DH algorithm dynamically balances two competing goals, i.e., maximizing the collection of task-related ground truth and the value of acquired environmental information. To achieve this, a hybrid optimization approach is adopted, integrating Multi-Objective Particle Swarm Optimization (MOPSO) with the Non-dominated Sorting Genetic Algorithm II (NSGA-II). Furthermore, a cooperative mechanism is established in which the environmental intelligence gathered by the UAV is used to refine the tasks assigned to ground-based participants. This synergy demonstrably improves the efficiency of task execution and reduces energy consumption, highlighting the practical value of the environmental data collected. Experimental results on real-world data demonstrate that the Co-UPP framework significantly outperforms conventional strategies in terms of overall system utility, task efficiency, and robustness.

Index Terms—Mobile CrowdSensing, UAV, Path Planning, Co-Optimization, Environmental Sensing, Task Allocation.

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I. Introduction

WITH the rapid advancement of mobile communication, sensor, and chip technologies, smart portable terminals such as smartphones, tablets, and wearable devices have become ubiquitous and increasingly powerful. Equipped with a variety of advanced sensors, these portable terminals are able to collect and transmit rich environmental information through robust network connections, which can significantly propel the development of Mobile CrowdSensing (MCS). MCS is a cost-effective paradigm for collaborative tasks, especially large-scale distributed data collection. MCS has garnered substantial attention from both industry and academia, finding widespread application in various domains, including urban traffic monitoring, environmental sensing, and public safety initiatives.

Recent advances in edge intelligence further enable real-time processing and coordination for large-scale sensing systems, including dynamically allocating users in collaborative large-small model settings [1], jointly optimizing resource and pricing for collaborative inference at the edge [2], and improving distributed hierarchical model training efficiency via adaptive aggregation structures [3]. Moreover, the environmental information of interest can extend beyond traffic states to wireless context maps (e.g., distributed radio map estimation and efficient transmission of radiomaps) [4], [5], where accurate channel estimation with low pilot overhead is also critical [6].

Despite its promise, existing MCS systems frequently face significant challenges, mainly due to allocation strategies and inefficient resource utilization. A critical underlying factor contributing to these inefficiencies is the pervasive lack of real-time environmental awareness, which hinders the system's ability to adapt to dynamic conditions and make informed decisions. Although Unmanned Aerial Vehicles (UAVs) have emerged as a promising technology offering novel possibilities for data collection and environmental monitoring, their integration into MCS frameworks remains limited. Accordingly, reliable control and coordination mechanisms are highly demanded [7]. Traditionally, UAVs' roles are frequently confined to sparse ground truth collection or static surveillance, leaving their broader, dynamic sensing capabilities largely

underutilized in enhancing overall MCS effectiveness.

To address the limitations mentioned above and unlock the full potential of UAVs in augmenting MCS operations, this paper proposes a Co-Optimized UAV Path Planning (Co-UPP) framework. Co-UPP is designed to significantly improve the effectiveness of MCS by integrating dynamic environmental sensing capabilities provided by UAVs, thus improving real-time awareness and optimizing ground worker task assignments.

The development of such a framework presents two primary challenges. First, the development of an effective UAV path planning strategy is at the research frontier. It must balance the need to maximize the collection of specific task-related ground truth data against the goal of maximizing the value of broader environmental information acquired during the flight. Achieving this balance is crucial to ensure that UAV operations are not only task-specific, but also contribute to a comprehensive understanding of the operational environment. Second, a robust cooperative mechanism is essential to seamlessly integrate the environmental intelligence gathered by the UAVs with the task allocation for ground workers. This synergy is vital to translate the enhanced environmental awareness into tangible improvements in task execution efficiency and resource management for the entire MCS system.

To tackle these challenges, Co-UPP introduces a novel approach with the following key contributions:

- 1) A multi-objective UAV path planning algorithm, termed the Evolutionary Fusion Dual-stage Hybrid (EF-DH) algorithm, is proposed as the core of the Co-UPP framework. The EF-DH algorithm is meticulously designed to dynamically balance two critical and often competing objectives, i.e., maximizing the collection of task-related ground truth data and maximizing the value of acquired environmental information.
- 2) To leverage the rich environmental intelligence gathered through the UAV's optimized path planning, a cooperative mechanism is proposed. This intelligence is then strategically utilized to refine and optimize task assignments for ground workers, creating a synergistic relationship between aerial and ground-based sensing assets. This cooperative approach demonstrably leads to higher task execution efficiency and significantly lower energy consumption across the MCS system, showcasing a practical and impactful application of the environmental data collected.
- 3) Comparison experiments conducted on both real-world datasets and simulated data rigorously verify the efficacy of the proposed Co-UPP framework. The results demonstrate that Co-UPP significantly outperforms conventional strategies in terms of overall system utility, task execution efficiency, and robust performance under various operational conditions.

The following section reviews existing relevant studies to elaborate the research gap addressed by Co-UPP.

II. Related Work

In this section, we review literature pertinent to the challenges addressed in this paper, structured into three areas. First, it examines task allocation strategies in MCS, highlighting their limitations. Second, it explores the role of UAVs in MCS, assessing their current applications. Finally, it investigates multi-objective optimization for UAV path planning. This review identifies the critical research gap that motivates the Co-UPP framework.

A. Task Allocation Strategies in MCS

Task allocation is a fundamental research issue in MCS, dictating the strategic assignment of crowd tasks to mobile workers. The primary research challenge is how to optimize these assignments against competing factors like maximizing spatio-temporal coverage, minimizing costs, and reducing task completion time. For instance, Zheng et al. [8] formulated system throughput and workers' travel distance as score terms in an objective function to find a flexible compromise between these competing factors. Other researchers focus on specific challenges, including dealing with unpopular tasks. For example, Zhen et al. [9] proposed a mechanism to bundle unpopular tasks with popular ones to ensure their completion.

To improve assignment robustness in realistic urban settings, recent studies have increasingly incorporated mobility forecasting and real-time route information. For example, trajectory prediction has been used to determine task assignments for candidate workers [10], and real-time route prediction further supports multi-objective task allocation in opportunistic mobile CrowdSensing [11]. In edge-cloud networks, task allocation can also be optimized by combining trajectory prediction with spatial partitioning mechanisms such as Voronoi diagrams [12]. In addition, dispatching-oriented optimization has been studied in related MEC scenarios, such as microservice request dispatching with game-theoretic modeling [13], which further motivates the need for principled assignment strategies under dynamic system states.

Methodologies have evolved from simple pull models, where workers self-select tasks, to more sophisticated platform-centric push models. Such push models have further advanced from static, offline scenarios to more realistic online models that must contend with the dynamic arrival of tasks and workers. A significant portion of research addresses the recruitment of workers for complex tasks, particularly for scenarios requiring multiple workers to collaborate. Recently, fairness-aware reinforcement learning has been adopted to support dynamic co-optimization in multi-space CrowdSensing task allocation [14].

The quality of data submitted by workers is another critical concern. To address this, Wang et al. [15] proposed a game-based quality control system that considers the costs of verification to ensure workers submit high-quality data. Reinforcement learning has also been leveraged to enhance game policy optimization in mobile CrowdSensing

[16]. Similarly, ensuring the freshness of data, or Age of Information (AoI), has been explored. Zhou et al. [17] utilized contract theory to design incentive mechanisms that motivate participants to perform high-frequency sampling for time-sensitive tasks. Beyond these, incentive and strategic interaction have been widely modeled via Stackelberg games and evolutionary games, including cooperative two-stage incentive mechanisms [18], three-party dynamic multi-strategy evolutionary games [19], and cross-region task allocation based on worker coalitions [20].

Despite this sophistication, a critical limitation persists in that these models operate without real-time awareness of the physical environment. While online frameworks are capable of reacting to changes in participant and task pools, they cannot proactively adapt to dynamic environmental conditions like physical obstructions or transient congestion. This creates a last-mile inefficiency problem, where an assignment that appears optimal on a map proves infeasible or inefficient in reality. A worker may discover an unexpected blockage only upon arrival, leading to wasted time, excess energy consumption, and potential task failure. This reveals a significant gap between algorithmic optimization and real-world operational effectiveness, which this work aims to address.

B. The Role of UAVs in Enhancing CrowdSensing

The integration of UAVs into MCS frameworks addresses the limitations of human-centric sensing, especially in areas that are inaccessible or hazardous for human participants. As summarized in the survey by Wang et al. [21], UAV-assisted sensing has seen significant development in recent years. Characterized by high agility and flexible deployment, UAVs serve as a powerful complement to ground-based workers.

In the literature, UAVs primarily fulfill several key functions. They could act as autonomous data collectors to replace or augment human workers. For example, Liu et al. [22] used a UAV as a data collection collaborator that is more inclined to fly to areas that ground workers cannot access. They also serve as data mules for ferrying information in areas with poor network infrastructure. Furthermore, they could function as data quality calibrators. In this vein, Chen et al. [23] employed UAVs to collect ground truth data, which is then used to test and calculate the trustworthiness of workers' data.

However, this paradigm confines UAVs to task-centric roles, treating them as just another type of worker or as auxiliary infrastructure. Much of the current research focuses on overcoming the operational constraints of UAVs. For instance, due to limited energy, Chen et al. [24] proposed a reinforcement learning approach that allows a UAV to adaptively switch between data collection and charging modes, while Chen et al. [25] used Unmanned Ground Vehicles (UGVs) to provide timely recharging, thereby maximizing the UAV's flight time. There are some other works that focus on the privacy concerns of UAVs in MCS. For example, Xie et al. [26] proposed a federated

learning framework to avoid centralizing privacy-sensitive data gathered by UAVs. Related privacy-preserving mechanisms in mobile crowdsourcing include incentive designs based on homomorphic encryption and edge computing [27]. More broadly, federated learning in open-world settings [28] and federated continual learning with generative replay [29], [30] provide promising directions for privacy-sensitive, adaptive intelligence at the edge.

In a nutshell, most existing studies restrict the role of UAVs to direct data collection, effectively treating them as auxiliary sensors comparable to ground-based workers. However, this perspective overlooks the critical capability of UAVs to acquire real-time environmental information, such as physical obstructions or crowd density, which is essential for robust task allocation. By decoupling UAV sensing from the decision-making process, current frameworks utilize the UAV solely to execute specific sensing tasks, failing to leverage its observations to refine task assignments for ground participants. This highlights a structural disconnect between aerial perception and ground-level decision-making, which the Co-UPP framework is designed to bridge.

C. Multi-Objective Optimization for UAV Path Planning

UAV path planning is inherently a Multi-Objective Optimization Problem (MOPP), as a feasible path must satisfy numerous, often conflicting, criteria. The goal is not a single optimal solution but a set of Pareto-optimal solutions, each representing a trade-off where one objective cannot be improved without degrading another.

Common objectives focus on path efficiency, safety, and data-centric goals. For example, Deng et al. [31] jointly optimized a UAV's hovering position, bandwidth allocation, and sensing data size to minimize overall latency for delay-sensitive tasks. A significant focus has been placed on minimizing the Age of Information (AoI). Both Liu et al. [22] and Chen et al. [25] proposed Deep Reinforcement Learning strategies for UAV path planning with the explicit goal of minimizing the global AoI. In another domain, Wu et al. [32] focused on cooperative search, where they jointly optimize flying trajectories and resource allocation to maximize the number of target discoveries and coverage rate. Related metaheuristic work has also examined tabu-search-based path planning in UAV-assisted edge computing systems [33], which supports the use of hybrid evolutionary search for UAV trajectory optimization.

However, a critical survey reveals that these objective functions are fundamentally UAV-centric. They optimize for the UAV's own performance, safety, or direct data output, but overlook the global optimality for the entire cooperative system. For instance, the shortest path for a UAV might bypass a vantage point from which it could gather critical intelligence about a traffic jam, information that could save ground workers significant time and energy. This points to the need for a new objective function that explicitly quantifies the value of

environmental intelligence as a resource to be leveraged to improve the operational efficiency of the entire system.

In summary, two imperative challenges are revealed from the related works. First, MCS task allocation is limited by a lack of real-time environmental awareness. Second, UAVs are often underutilized, primarily serving as task executors rather than active environmental sensors. This gap is particularly evident in research that has not yet fully addressed the co-optimization problem, where a UAV's path is dynamically planned to balance its own sensing tasks with gathering valuable environmental intelligence for other agents. As a result, current systems rarely use this intelligence to refine task allocation for ground workers in real time. The proposed Co-UPP framework aims to fill this gap by creating a genuinely cooperative mechanism that elevates the UAV from a worker to an active environmental sensor, improving overall system utility, task efficiency, and energy conservation.

With these research limitations clarified, we next formalize the system model and formulate the optimization problem.

III. System Model and Problem Statement

In this paper, we address a two-stage optimization problem for a UAV-worker collaborative system designed for efficient task completion in a dynamic urban environment. The first stage involves planning optimal flight paths for a fleet of UAVs to survey tasks and the environment. The second stage utilizes the information gathered by the UAVs to assign tasks to a team of ground workers. Our objective is to find Pareto-optimal solutions for both stages, balancing multiple conflicting objectives. The key notations used in our model are summarized in Table I.

A. Problem Description

We consider a static sensing scenario on a square map of size $L \times L$. The environment is an orthogonal road network with N vertical and M horizontal roads, where intersections split roads into atomic segments. Let $G = (V, E)$ be the road graph with vertices V representing intersections and edges E representing atomic road segments. A set of sensing tasks T is distributed near roads, a fleet of UAVs U departs from a designated depot, and a set of ground workers W is initialized on the road network.

Definition 1 (UAV): A UAV $u_i \in U$ operates with a fixed sensing radius $\rho > 0$ and executes a travel path in the 2D plane starting from a designated depot. A task is regarded as sensed by u_i if its location is within distance ρ of the UAV's path.

Definition 2 (Worker): A worker $w_i \in W$ is located on the road network G and is described by a position on R and ability metadata (ability type and ability level). Worker movements are restricted to paths on G when traveling between locations and executing assigned tasks.

Definition 3 (Sensing Task): A sensing task $t_j \in T$ is specified by its location p_j , a reward parameter, and

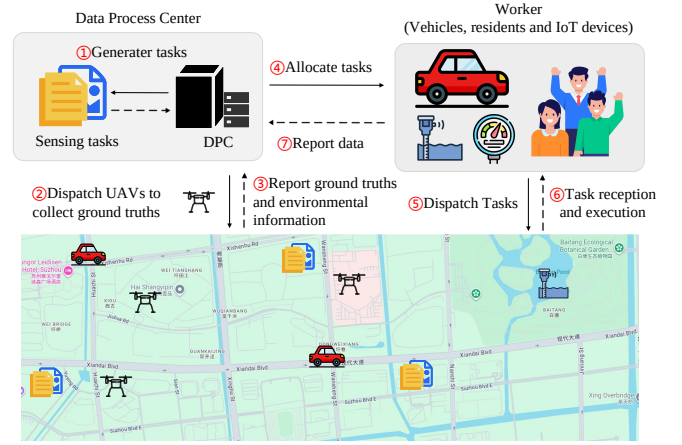


Fig. 1. Data collection system model.

auxiliary attributes such as task type, difficulty, and nominal time requirement.

Definition 4 (Road Segment): The operational environment is defined by a set of road segments E . Each road segment $e_i \in E$ is characterized by its start and end coordinates, a congestion level $c_i \in [0, 1]$, and an environmental value v_i . The value v_i represents the importance of gathering information about that specific road segment.

Definition 5 (Travel Path): A travel path is an ordered sequence of waypoints connected by straight segments.

- UAV path: a polyline in the plane starting at a depot, used for sensing within radius ρ .
- Worker path: a sequence of edges on the road graph G connecting the worker's current position to one or more task locations.

In summary, given the road network with per-segment congestion values, the distributed tasks, the UAV fleet, and ground workers, the problem is to jointly plan UAV routes and assign/sequence worker travel on the road graph to balance sensing utility (task coverage and rewards), environmental information gain, and operational costs (travel distance and completion time), under practical feasibility constraints. The solution should coordinate information gathering and task execution to achieve effective coverage and timely completion. The detailed multi-objective formulation and solution approach are presented in the next section.

B. System Model

The system comprises a set of tasks distributed across a geographical area, a fleet of UAVs stationed at a central depot, and a team of ground workers positioned throughout the area. The environment is modeled as a road network with varying levels of congestion. The process flow is illustrated in Fig. 1.

- 1) **UAV Path Planning:** The system first solves a multi-objective path planning problem for the UAV fleet. The UAVs start from a central depot $(0, 0)$, visit a subset of task locations, and return to the depot. The

TABLE I
Summary of Notations

Symbol	Meaning
T	Set of tasks
t_j	An individual task
U	Fleet of UAVs
u_i	An individual UAV
W	Team of workers
w_i	An individual worker
E	Set of road segments
e_i	A single road segment
M	Number of UAVs
N	Number of workers
P	Set of all UAV paths
p_i	Path for UAV i
$Y = \{y_{ij}\}$	UAV visit indicator matrix
$X = \{x_{ij}\}$	Worker-task assignment matrix
$\mathcal{R} = \{\xi_i\}$	Set of worker routes (route ξ_i for worker i)
$\mathcal{T}_i$	Set of tasks assigned to worker w_i
S	Road status information collected by UAVs
B_i	Battery budget of UAV i
$E_i(p_i)$	Energy consumed by UAV i on path p_i
f_{rew}	UAV objective: reward
f_{env}	UAV objective: environment value
f_{dist}	UAV objective: distance
f_{time}	Worker objective: makespan
f_{val}	Worker objective: total value

goals are to maximize the collected task rewards, maximize the value of the environmental information gathered from flying over roads, and minimize the total flight distance.

- 2) Information Relay: The optimal UAV path plan yields critical real-time information about the road network, specifically, which road segments were surveyed. This creates an updated map with status of roads.
- 3) Worker Task Assignment: Using the map with status of roads from the UAVs, the system then solves a multi-objective task assignment problem for the ground workers. The goals are to minimize the overall time required for all tasks to be completed (makespan) and maximize the total value generated by the workers.
- 4) Battery Feasibility: Each UAV path must satisfy an energy-budget constraint so that planned routes are executable under finite onboard battery.

1) UAV Path Planning Objectives: Let $P = \{p_1, p_2, \dots, p_M\}$ be the set of paths for all UAVs. The path planning problem is formulated with three objectives. These objectives are not independent. Task reward captures direct sensing utility, environmental value captures the indirect decision-making benefit of surveying informative road segments, and path distance measures the operational cost paid to obtain both forms of information. The Pareto formulation therefore keeps the UAV from over-focusing on any single aspect and allows the planner to trade off direct reward, environmental intelligence, and flight efficiency.

- 1) Maximize Task Reward (f_{rew}): The sum of rewards from all unique tasks whose locations are visited by

any UAV.

$$\max f_{rew}(P) = \sum_{t_j \in T_{visited}} r_j. \quad (1)$$

where $T_{visited}$ is the set of unique tasks on all paths in P .

- 2) Maximize Environmental Value (f_{env}): The sum of values from all unique road segments traversed by the UAV fleet. In our setting, v_i reflects how informative a segment is for downstream task scheduling, so a route that covers more valuable segments can support better worker-side decisions even if it does not directly visit more tasks.

$$\max f_{env}(P) = \sum_{e_i \in E_{traversed}} v_i. \quad (2)$$

where $E_{traversed}$ is the set of unique road segments covered by paths in P .

- 3) Minimize Total Distance (f_{dist}): The sum of the Euclidean lengths of all UAV paths.

$$\min f_{dist}(P) = \sum_{i=1}^M \text{length}(p_i). \quad (3)$$

2) Worker Task Assignment Objectives: Let $X = \{x_{ij}\}$ be an assignment matrix where $x_{ij} = 1$ signifies that task t_j is assigned to worker w_i . Let S be the road status information derived from the selected UAV path set P^* . The worker assignment problem has two objectives:

- 1) Minimize Makespan (f_{time}): The maximum completion time among all workers. This is the time when the last task is completed, and it directly measures whether the task set can be finished without creating severe load imbalance.

$$\min f_{time}(X, S) = \max_{i \in W} \{T_i(\xi_i, S)\}. \quad (4)$$

where $T_i(\xi_i, S)$ is the total time worker w_i spends traveling and executing assigned tasks along route ξ_i , with travel time calculated using road status S .

- 2) Maximize Total Value (f_{val}): The sum of the value generated by all workers. The value of a task is a function of its base reward, the assigned worker's ability, and the quality of information from the UAV survey. This objective captures the practical scheduling benefit of using UAV-collected road information to improve downstream execution.

$$\max f_{val}(X, S) = \sum_{i \in W} V_i(\xi_i, S). \quad (5)$$

where $V_i(\xi_i, S)$ denotes the total value generated by worker w_i along route ξ_i under road-status information S . In our model, this value is composed of the rewards of the tasks assigned to worker w_i and an additional bonus contributed by the road segments surveyed by the UAVs:

$$V_i(\xi_i, S) = \sum_{t_j \in \mathcal{T}_i} r_j + \eta \sum_{e \in \xi_i \cap E_{surveyed}(S)} v_e \quad (6)$$

where:

- $\mathcal{T}_i$ is the set of tasks assigned to worker w_i
- r_j is the base reward of task t_j
- ξ_i denotes the travel path of worker w_i
- $E_{\text{surveyed}}(S)$ is the set of road segments surveyed by UAVs and recorded in S
- v_e is the environmental value of road segment e
- η is a weighting coefficient that controls how strongly UAV-collected road information contributes to worker-side value

C. Problem Statement

Based on the above model, we define two interconnected multi-objective optimization problems.

First, the UAV Multi-Objective Path Planning (MOPP) problem is to find a set of paths P that optimizes the defined UAV objectives.

Let $P = \{p_i\}_{i \in U}$ be the set of UAV paths and $Y = \{y_{ij}\}$ be the binary visit matrix where $y_{ij} = 1$ means UAV i visits task j . The objectives are to maximize task sensing utility and environmental information gain while to minimize flight distance:

$$\mathbf{F}_{\text{UAV}}(P) = \begin{cases} \max & f_{\text{rew}}(P), \\ \max & f_{\text{env}}(P), \\ \min & f_{\text{dist}}(P). \end{cases} \quad (7)$$

Subject to assignment and feasibility constraints:

$$Y = \{y_{ij} \mid y_{ij} \in \{0, 1\}, 1 \leq i \leq |U|, 1 \leq j \leq |T|\}, \quad (8)$$

$$0 \leq \sum_{i=1}^{|U|} y_{ij} \leq 1, \quad 1 \leq j \leq |T|, \quad (9)$$

$$E_i(p_i) \leq B_i, \quad \forall i \in U. \quad (10)$$

The output is a Pareto set of non-dominated P . Subsequently, one solution P^* is selected to produce the derived road status S^* .

Second, the Worker Multi-Objective Task Assignment (MOTA) problem is to find a set of assignment that optimizes the defined Worker objectives.

Given S^* , let $X = \{x_{ij}\}$ be the assignment matrix where $x_{ij} \in \{0, 1\}$ indicates whether worker $i \in W$ is assigned task $j \in T$. This assignment together with S^* induces a set of worker paths $\mathcal{R} = \{\xi_i\}_{i \in W}$ on the road graph. The objectives are to minimize overall completion time and maximize worker-side value:

$$\mathbf{F}_{\text{WRK}}(X) = \begin{cases} \min & f_{\text{time}}(X, S^*) = \max_{i \in W} T_i(\xi_i, S^*), \\ \max & f_{\text{val}}(X, S^*) = \sum_{i \in W} V_i(\xi_i, S^*). \end{cases} \quad (11)$$

Subject to assignment and feasibility constraints:

$$X = \{x_{ij} \mid x_{ij} \in \{0, 1\}, 1 \leq i \leq |W|, 1 \leq j \leq |T|\}, \quad (12)$$

$$0 \leq \sum_{i=1}^{|W|} x_{ij} \leq 1, \quad 1 \leq j \leq |T|. \quad (13)$$

The solution yields a Pareto set of non-dominated assignments. The resulting heterogeneous planning-and-assignment problem is a multi-objective integer (and path) optimization problem and is NP-hard. In practice, the number of tasks and workers is large, making exact optimization intractable within polynomial time. Therefore, we adopt heuristic and metaheuristic algorithms to obtain high-quality suboptimal solutions, with details presented in the next section.

IV. System Design

To solve the MOPP and MOTA problems, we propose an Evolutionary Fusion Dual-stage Hybrid framework called EF-DH. The core idea of EF-DH is to evolutionarily fuse the global exploration capability of MOPSO in the early search stage with the robust exploitation and convergence capability of NSGA-II in the later stage. This dual-stage hybrid evolution is applied sequentially to solve the MOPP problem first and the MOTA problem subsequently. Fig. 2 presents the modular diagram of the EF-DH framework, depicting the stepwise execution flow of its key functional modules.

A. Solution Encoding

For both the MOPP and MOTA problems, the solution, a particle in MOPSO or a chromosome in NSGA-II, is encoded as an integer array of length m , where m is the total number of tasks.

- For the MOPP problem, the j -th element of the array contains the ID of the UAV (u_i) assigned to survey task t_j .
- For the MOTA problem, the j -th element of the array contains the ID of the worker (w_i) assigned to execute task t_j .

B. MOPSO Phase for Global Exploration

The algorithm begins with the MOPSO phase, which runs for the first half of the total iterations. This phase is designed to broadly search the solution space and identify promising regions.

- 1) Initialization: A swarm of particles is initialized with random assignments. Each particle has an associated velocity vector, initialized to zero. An external archive is created to store the non-dominated solutions identified during the search process.
- 2) Evaluation: The fitness of each particle is evaluated based on the corresponding multi-objective functions (either for MOPP or MOTA).
- 3) Personal Best Update: For each particle, its current position is compared with its personal best position ($pbest$). The new position replaces $pbest$ if it dominates the old one.
- 4) Archive Update: Each particle's current position is compared against the solutions in the external archive. If the particle's solution dominates any archive members, those members are removed. If no

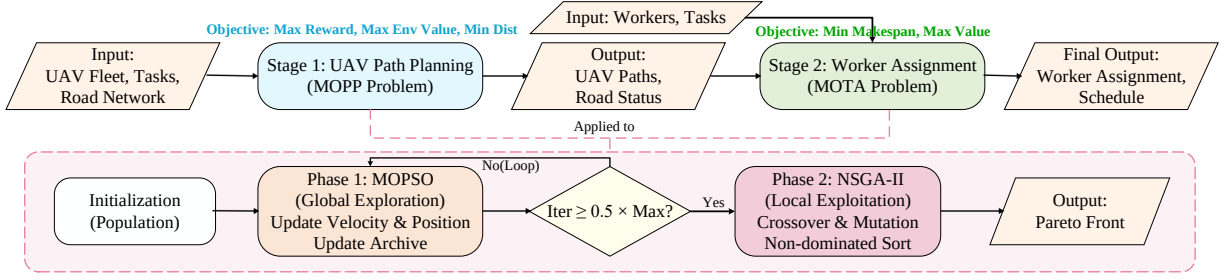


Fig. 2. Flowchart of EF-DH Framework.

archive member dominates the particle's solution, it is added to the archive. The archive has a fixed size limit.

- 5) Velocity and Position Update: A leader is selected randomly from the archive. The velocity and position of each particle i are updated according to the following equations:

$$v_i(t+1) = w \cdot v_i(t) + c_1 \cdot r_1 \cdot (pbest_i - x_i(t)) + c_2 \cdot r_2 \cdot (leader - x_i(t)) \quad (14)$$

$$x_i(t+1) = \text{round}(x_i(t) + v_i(t+1)) \quad (15)$$

where w is the inertia weight, c_1 and c_2 are acceleration coefficients, and r_1, r_2 are random numbers in $[0, 1]$. The updated position is clipped to ensure valid UAV/worker IDs.

C. NSGA-II Phase for Local Exploitation and Convergence

After the MOPSO phase concludes, the algorithm transitions to the NSGA-II phase. The final population from the MOPSO phase serves as the initial population for NSGA-II. This phase focuses on refining the solutions and achieving a well-distributed Pareto front.

- 1) Offspring Generation: An offspring population of the same size as the parent population is created using genetic operators.
 - Crossover: Pairs of parents are selected, and a single-point crossover is applied to create two children.
 - Mutation: A mutation operator is applied to the offspring. With a small probability, a randomly selected task in a child's assignment array is reassigned to a new random UAV/worker.
- 2) Elitist Selection: The parent and offspring populations are combined. The new population for the next generation is selected from this combined set based on non-domination rank and crowding distance.
 - Fast Non-Dominated Sort: All solutions in the combined population are ranked and sorted into different non-domination fronts ($F_1, F_2, \dots$). Solutions in F_1 are the best, followed by F_2 , and so on.
 - Crowding Distance: Within each front, a crowding distance is calculated for each solution.

This metric estimates the density of solutions surrounding a particular solution, promoting diversity in the population.

- New Population Construction: The new population is filled by adding all solutions from front F_1 , then F_2 , and so on, until the population size would be exceeded. For the last front to be added, solutions are selected based on their crowding distance in descending order until the population is full.

This process repeats for the remaining iterations, ensuring that the best solutions are preserved and refined over time.

D. The EF-DH Collaborative Scheme

The complete EF-DH scheme for solving the UAV-worker problem is detailed in Algorithm 1. The process begins by applying the hybrid algorithm to the MOPP problem. Once the UAV Pareto front is obtained, a single best solution is chosen by finding the point with the minimum Euclidean distance to an ideal point. This best UAV solution is then used to generate the road status information required for the worker assignment stage. The reason for placing UAV optimization first is that the worker-side objective depends on road-status information derived from the UAV stage. Without this information, the assignment stage would be forced to operate on an incomplete view of the environment. Finally, the hybrid algorithm is applied again, this time to solve the MOTA problem using the UAV-derived road data. The final output is a Pareto front of efficient worker assignments.

E. Mechanism Interpretation of UAV-to-Worker Cooperation

To make the cooperative effect explicit, we describe the information-to-decision chain used in Co-UPP:

- 1) Observation abstraction. The selected UAV route set P^* induces a road-status map S^* by marking traversed segments and associated congestion-relevant context.
- 2) Cost update. Worker-side travel-time evaluation is recalculated on the road graph conditioned on S^* , which modifies route costs $T_i(\xi_i, S^*)$ in Eq. (11).
- 3) Assignment reshaping. Because route costs change after the update, the optimizer tends to reassign

Algorithm 1 EF-DH Collaborative Scheme

Input: Task set T , Road network E , Worker team W , UAV fleet U

Output: A_{pareto}

- 1: Stage 1: UAV Path Planning
 - 2: Initialize population Pop_{uav} for the MOPP problem.
 - 3: Run Hybrid MOPSO-NSGA-II on Pop_{uav} using objectives from Eqs. (7), (8), (9), and battery constraint Eq. (10).
 - 4: Obtain UAV Pareto front P_{pareto} .
 - 5: Select best solution P^* from P_{pareto} based on distance to ideal point.
 - 6: Generate road status map S^* from P^* .
 - 7: Stage 2: Worker Task Assignment
 - 8: Initialize population Pop_{worker} for the MOTA problem.
 - 9: Run Hybrid MOPSO-NSGA-II on Pop_{worker} using objectives from Eqs. (11), (12), (13).
 - 10: Obtain worker Pareto front A_{pareto} .
 - 11: return A_{pareto}
-

tasks away from high-friction regions and rebalance workloads across workers.

- 4) Metric-level impact. This pathway primarily affects worker makespan and worker value, while UAV-side objectives are determined in Stage 1 under the battery-feasible path set.

This interpretation clarifies that the gain of the two-stage scheme is not only from the hybrid optimizer itself, but also from the coupling interface $P^* \rightarrow S^* \rightarrow X$. In practical terms, the UAV does not merely collect data for its own path quality. It exposes hidden road constraints that shape the worker allocation landscape and reduce the chance of assigning tasks through congested or inefficient routes.

V. Experiment and Analysis

A. Objectives and Research Questions

This section empirically evaluates the proposed EF-DH framework. We address the following research questions (RQs):

- RQ1: Does EF-DH improve overall system performance in terms of task completion time and information value compared with single-stage and single-algorithm baselines?
- RQ2: How does EF-DH maintain a balance among UAV-level exploration objectives including reward maximization, distance minimization and environmental coverage as the task scale increases?
- RQ3: What impact does the dual-stage collaborative mechanism, consisting of UAV reconnaissance followed by informed worker assignment, exert on the temporal efficiency and value outcomes of the worker side?

B. Datasets and Preprocessing

We employ three real taxi trajectory datasets covering heterogeneous urban morphologies: T-Drive Taxi Trajectory Dataset (Beijing, China, 2010, Microsoft Research Asia, [34]–[36]), Porto Taxi Dataset (Porto, Portugal, 2013-2014, ECML/PKDD 2015 Challenge, [37]), and Rome Taxi Mobility Dataset (Rome, Italy, 2016, [38]). Raw GPS traces (longitude, latitude, timestamp) are map-aligned and discretized into a road-segment set. For each of them we additionally rasterize the spatial extent into a 100×100 grid and accumulate per-cell traversal counts. Each GPS point is filtered, projected into (lon, lat) bounds, and mapped to a road segment. Per-segment empirical congestion c_e is derived via normalized visit frequency (placeholder scaling: min-max normalized).

C. Generations of Tasks, UAVs and Workers

For each task scale (number of tasks $|T| \in \{10, 20, \dots, 300\}$), task locations are sampled near road segments proportional to v_e (bias factor = 1, i.e., tasks are uniformly sampled over road segments while v_e is encoded in the segment value) to emulate realistic sensing demand clustering.

Task attributes (base reward, difficulty level, and nominal service time) follow the distributions: reward $\sim \mathcal{U}[0, 1]$, difficulty level $d \in \{0, 1, \dots, 10\}$, and the service time is modeled as

$$t^{\text{srv}} = f(d) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1). \quad (16)$$

Worker abilities are stratified into levels $\{0, 1, \dots, 10\}$ with a productivity multiplier $m_a = 1$ (i.e., the ability value directly represents the worker's productivity). The UAV sensing radius is set to $\rho = 5$, and the fleet size and worker population are fixed unless explicitly varied (default: $|U| = 5$, $|W| = 8$). The depot is located at $(0, 0)$ in the normalized coordinate system.

We compare:

- EF-DH (proposed): A hybrid framework integrating MOPSO followed by NSGA-II in each stage, featuring a two-stage coupling structure where UAV planning is followed by worker allocation.
- MOPSO-DH: A variant employing pure MOPSO in each stage while retaining the dual-stage coupling mechanism.
- NSGA-II-DH: A variant employing pure NSGA-II in each stage, similarly retaining the dual-stage coupling mechanism.
- EF: An evolutionary fusion approach without dual-stage coupling. It operates as a single-stage abstraction where worker assignment remains uninformed by UAV-gathered environmental intelligence.
- GREEDY: A value-over-cost heuristic where tasks are first sorted by reward in descending order and then assigned to the nearest UAV in terms of travel distance, i.e., a reward-per-distance greedy rule, followed by a greedy worker assignment based on the marginal task value and travel time.

- EHTA(SA) [39]: A simulated-annealing-based variant adapted from environment-cost-based heterogeneous task allocation, using a geometric temperature schedule $T_t = T_0 \cdot \lambda^t$ with $T_0 = 800$ and $\lambda = 0.98$ in our experiments.

D. Evaluation Metrics

We report six primary metrics (plotted against task count):

- UAV Reward: Total unique task reward sensed, defined as $f_{\text{rew}} = \sum_{j \in T_{\text{sensed}}} r_j$, where T_{sensed} is the set of tasks covered by at least one UAV and r_j is the reward of task j .
- UAV Total Distance: $\sum_i \text{length}(p_i) = \sum_{i \in U} \sum_{(u,v) \in p_i} \|u - v\|_2$, i.e., the sum of Euclidean lengths of all UAV routes p_i (including depot-task and task-depot legs).
- Total Time: Algorithmic runtime, i.e., the wall-clock time consumed by the whole procedure, defined as $\text{TotalTime} = t_{\text{worker_end}} - t_{\text{start}}$, which includes both the UAV routing stage and the worker assignment stage.
- Worker Maximum Time: Makespan $f_{\text{time}} = \max_{i \in W} T_i$, where T_i is the total travel time plus execution time of worker i .
- Worker Total Time: $\sum_i \text{Time}_{w_i} = \sum_{i \in W} T_i$, i.e., the sum of completion times over all workers.
- Worker Total Value: $f_{\text{val}} = \sum_{i \in W} \sum_{j \in \mathcal{T}_i} v_{i,j}$, where $\mathcal{T}_i$ is the task set assigned to worker i and $v_{i,j}$ is the effective value of worker i performing task j (task reward plus ability and information bonuses).

E. Result Analysis

All algorithms are implemented in Python 3.13.2. Experiments run on an Intel i5-12490F CPU (32 GB RAM) and NVIDIA 2080 Ti GPU (11 GB). Random seed is fixed to 42 unless otherwise stated. Population size is set to 50. Maximum iterations are 200 per stage. The hybrid switch point for EF-DH occurs at iteration ratio $\tau = 0.5$. Archive size in MOPSO period is 100. Inertia weight w linearly decays from 0.9 to 0.4. Cognitive/social coefficients are $(c_1, c_2) = (1.5, 1.5)$. NSGA-II uses single-point crossover with a rate of 0.9, and the mutation probability per gene is 0.05.

To systematically address the three research questions above, we conduct six targeted experiments focusing on the metrics of Total Running Time, UAV Reward, UAV Total Distance, Worker Maximum Time, Worker Total Time, and Worker Total Value. Experimental results and comparative analysis on three real-world datasets are detailed below.

Across all three datasets, EF-DH consistently maintains a competitive computational profile. While naturally slower than the heuristic Greedy approach, EF-DH reduces runtime by 49.0%–84.5% compared to NSGA-II-DH and 9.97%–23.7% compared to EHTA(SA). Its efficiency advantage over MOPSO-DH becomes more pronounced

as the dataset complexity increases, ranging from a 2.8% slight lag in Beijing to a 21.6% lead in Porto. Overall, the running time results indicate that the proposed hybrid strategy improves solution quality without introducing prohibitive computational overhead.

In terms of UAV Reward as shown in Fig. 4, the proposed EF-DH algorithm demonstrated consistently superior performance across all three datasets. When compared to the baseline methods, the EF-DH algorithm achieved an average reward that was at least 29% higher than Greedy and 4% to 6% higher than MOPSO-DH. Furthermore, it significantly outperformed NSGA-II-DH by 49.07% and yielded a 3.2% to 5% higher reward than EHTA(SA). The main takeaway is that EF-DH preserves direct sensing utility while still allowing the UAV to explore informative road segments.

This performance advantage stems from EF-DH’s dual-stage hybrid mechanism, where the MOPSO-based early stage ensures sufficient global exploration of high-reward tasks and the NSGA-II-based later stage refines the UAV path to converge to optimal solutions. This effectively addresses the exploration-exploitation imbalance of single-algorithm baselines, particularly in large-scale task scenarios.

When evaluating the UAV Total Distance across all three datasets, the proposed EF-DH algorithm consistently achieved superior optimization results. As shown in Fig.5, The algorithm successfully reduced the total travel distance by over 4% compared to Greedy and by more than 8.3% compared to MOPSO-DH. Furthermore, it demonstrated an improvement of approximately 5% over NSGA-II-DH and yielded a significant 20.4% distance reduction when compared to EHTA(SA).

Combined with the UAV reward results, this confirms that EF-DH could simultaneously maximize reward acquisition and minimize travel distance, a key requirement for balancing multiple UAV-level objectives including reward, distance and coverage as stated in RQ2.

Worker Maximum Time is a critical performance indicator for this problem, as it defines the overall makespan or the final completion time of all tasks. The proposed EF-DH algorithm demonstrated exceptional efficiency in minimizing this metric across all three datasets, as shown in Fig.6. This outcome directly validates the effectiveness of the dual-stage cooperation between UAV reconnaissance and worker assignment corresponding to RQ3. The result is especially important because it shows that environmental sensing contributes not only to better routes, but also to a shorter critical path on the worker side.

The most significant performance improvement was observed in comparison with the Greedy algorithm. The Greedy algorithm consistently yielded a makespan that was 513% to 800% longer than that of EF-DH. Furthermore, EF-DH substantially reduced the completion time relative to all other benchmark methods. It was 32% to 44% faster than MOPSO, 29% to 42% faster than NSGA-II, and at least 29% faster than EF, a single-stage variant that excludes the UAV reconnaissance module. EF-DH

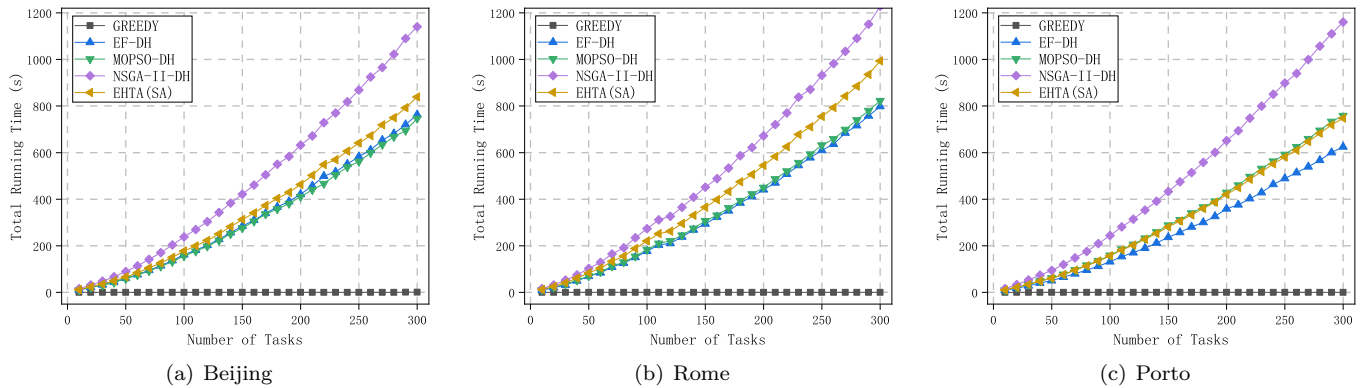


Fig. 3. Performance of total running time under different algorithms on three datasets with varying numbers of tasks.

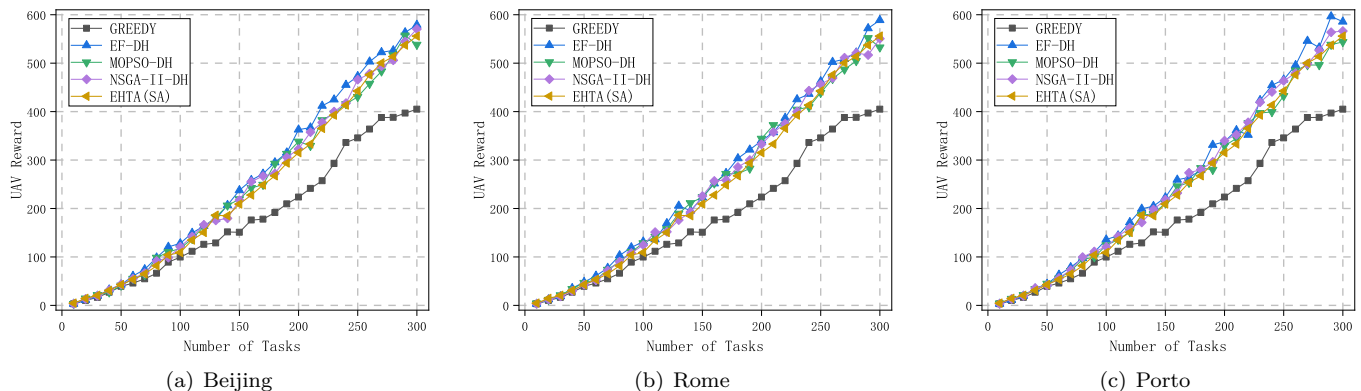


Fig. 4. Performance of UAV reward under different algorithms on three datasets with varying numbers of tasks.

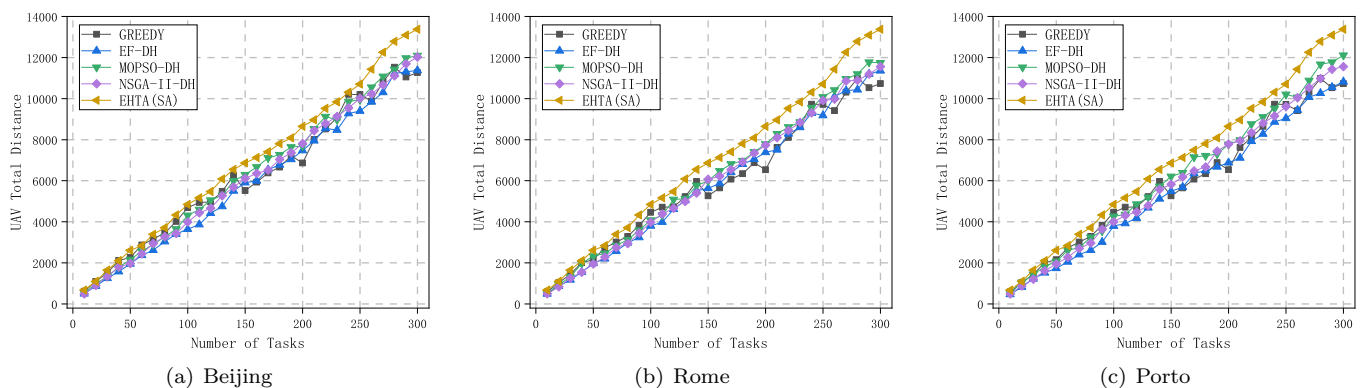


Fig. 5. Performance of UAV total distance under different algorithms on three datasets with varying numbers of tasks.

also maintained a notable advantage over EHTA(SA), achieving a reduction in maximum time by approximately 7% to 19%.

This striking reduction in makespan could be attributed to EF-DH's integration of UAV-derived environmental intelligence into worker task assignment. The UAV's reconnaissance results enable more rational task partitioning and routing, thereby avoiding bottlenecks in worker scheduling. Taken together, the makespan results indicate that the collaborative mechanism is the main driver of worker-side acceleration.

The Worker Total Time, representing the cumulative

operational time for all workers, was also analyzed. The proposed EF-DH algorithm demonstrated significant efficiency gains over most baseline methods as shown in Fig. 7. This metric complements makespan by showing whether improvements are limited to the slowest worker or extend to the entire workforce.

Specifically, it achieved a reduction in total time of 39% or more compared to NSGA-II, and at least 33% compared to both MOPSO and EF. An advantage was also shown over EHTA(SA), with an average reduction of approximately 6.6%. The only exception was the Greedy algorithm, against which EF-DH recorded an increase in

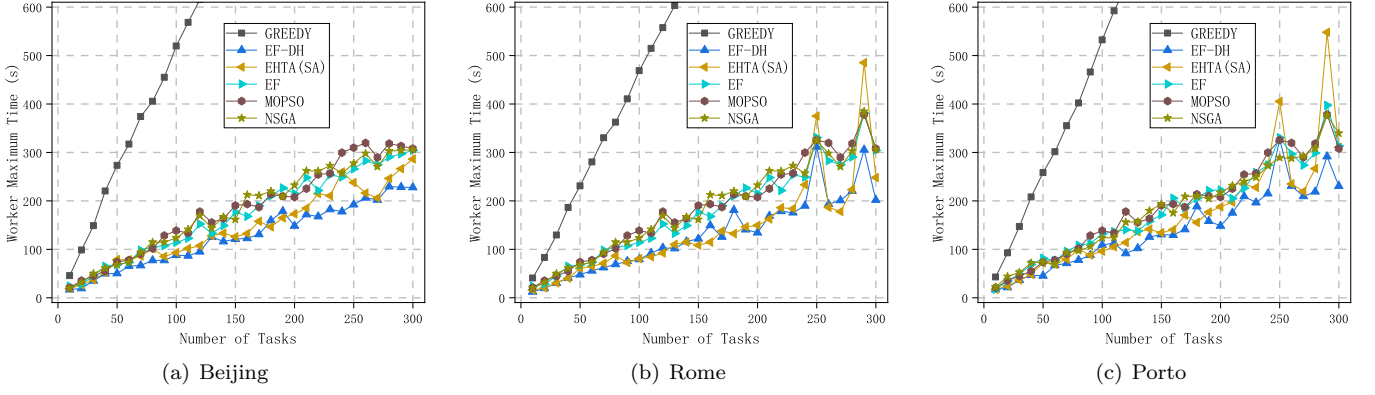


Fig. 6. Performance of worker maximum completion time under different algorithms on three datasets with varying numbers of tasks.

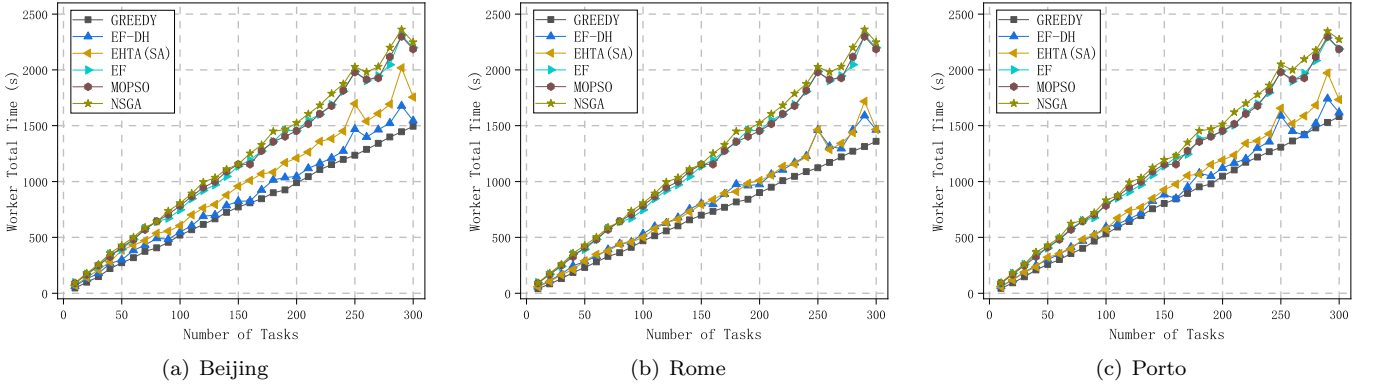


Fig. 7. Performance of worker total completion time under different algorithms on three datasets with varying numbers of tasks.

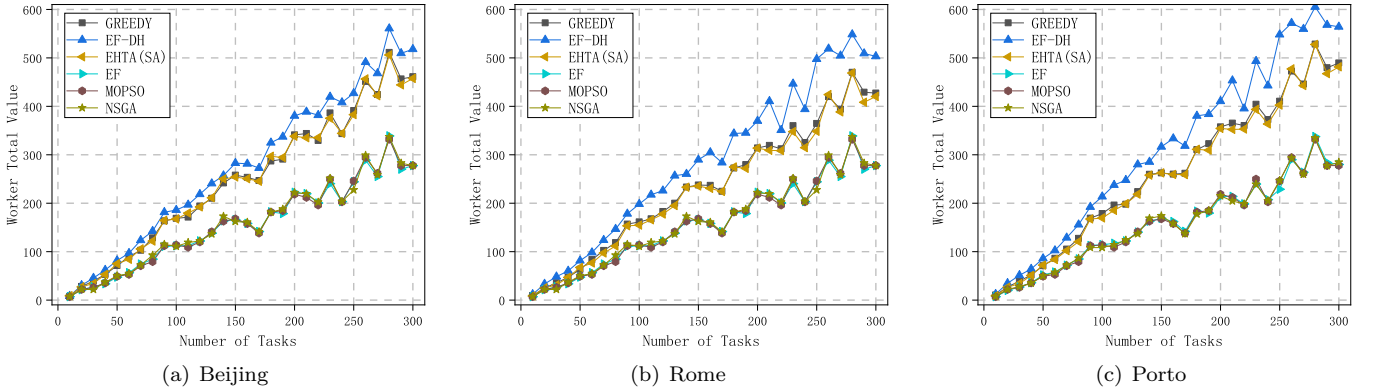


Fig. 8. Performance of worker total value under different algorithms on three datasets with varying numbers of tasks.

total worker time of approximately 9%. This discrepancy is attributed to the Greedy algorithm's strategy, which focuses on employing a minimal number of workers to fulfill the tasks, thereby reducing the cumulative time sum. Even so, the Greedy method sacrifices task value and makespan, so its advantage here reflects a narrower optimization target rather than better overall system performance.

As illustrated in Fig. 8, the Worker Total Value metric further confirms overall system efficiency. The proposed EF-DH algorithm maintains leading performance, which proves UAV-sensed road data is not merely supplementary

metadata and delivers tangible improvements to downstream task utility.

Across the three datasets, it achieved a substantially higher total value, recording improvements of 43% or more against both EF and MOPSO, and at least 42% against NSGA-II. Furthermore, it also consistently outperformed the other baselines, securing a value 10% to 18% higher than the Greedy algorithm and approximately 10% to 20% higher than EHTA(SA).

Taken together, these analyses of Worker Total Time and Worker Total Value collectively confirm that the dual-stage cooperation mechanism yields tangible improve-

TABLE II

Overall comparison on three datasets using mean values aggregated over task scales. Arrows indicate whether larger ($\uparrow$) or smaller ($\downarrow$) is better.

Dataset	Metric	GREEDY	EF-DH (Ours)	MOPSO- DH	NSGA-II- DH	EHTA(SA)	EF	MOPSO	NSGA
Beijing	Total Running Time $\downarrow$	0.26	325.88	316.6	485.78	358.47	6.43	2.92	6.52
	UAV Reward $\uparrow$	183.6	261.77	246.54	248.5	242.25	N/A	N/A	N/A
	UAV Total Distance $\downarrow$	6219.83	5906.77	6355.88	6182.75	6958.7	N/A	N/A	N/A
	Worker Max Time $\downarrow$	779.4	127.07	130.17	133.71	145.08	171.8	182.98	180.81
	Worker Total Time $\downarrow$	779.1	852.51	853.84	898.54	969.82	1169.02	1172.49	1219.91
Rome	Worker Total Value $\uparrow$	247.99	277.29	245.81	248.14	246.99	156.45	156.54	158.1
	Total Running Time $\downarrow$	0.15	341.98	352.21	521.88	423.06	7.92	3.68	8.12
	UAV Reward $\uparrow$	183.6	259.49	248.48	249.35	242.25	N/A	N/A	N/A
	UAV Total Distance $\downarrow$	5923.64	5850.12	6214.86	6027.2	6958.7	N/A	N/A	N/A
	Worker Max Time $\downarrow$	705.38	130.08	126.44	125.91	139.72	176.76	185.68	185.07
Porto	Worker Total Time $\downarrow$	705.38	808.12	772.06	811.81	808.65	1169.02	1172.49	1219.91
	Worker Total Value $\uparrow$	232.31	285.23	231.27	231.27	227.59	156.45	156.54	158.1
	Total Running Time $\downarrow$	0.15	270.97	329.56	500.02	323.31	7.9	3.59	8.14
	UAV Reward $\uparrow$	183.6	260.67	244.42	252.32	242.43	N/A	N/A	N/A
	UAV Total Distance $\downarrow$	5923.64	5584.08	6253.49	5987.83	6961.33	N/A	N/A	N/A
Porto	Worker Max Time $\downarrow$	814.86	140.26	141.05	145.82	166.94	180.95	185.68	182.08
	Worker Total Time $\downarrow$	814.86	879.4	890.72	938.14	938.81	1171	1172.49	1226.35
	Worker Total Value $\uparrow$	261.43	314.38	259.42	259.57	256.95	157.09	156.54	156.69

ments in both worker-side performance corresponding to RQ3 and the overall system efficiency associated with RQ1. The results also suggest that the benefit is strongest when tasks are distributed across segments with heterogeneous congestion and road-condition uncertainty.

To further provide a unified and quantitative comparison across all metrics and datasets, Table II presents the mean performance of all algorithms aggregated over different task scales.

From a global perspective, the proposed EF-DH algorithm consistently achieves the best or near-best performance across the majority of evaluation metrics. In particular, it attains the highest UAV reward and worker total value while maintaining clear advantages in UAV travel distance and worker completion efficiency. Although the Greedy algorithm exhibits lower total worker time and negligible running time due to its heuristic nature, it suffers from significantly inferior performance in reward, makespan, and overall system value. In contrast, EF-DH effectively balances multiple conflicting objectives, demonstrating strong robustness and superiority in complex multi-objective UAV-assisted CrowdSensing scenarios. The comparative tables therefore support the paper's central claim that collaborative sensing is more effective than isolated UAV search or isolated worker assignment. These results further corroborate the effectiveness of the proposed dual-stage hybrid mechanism in achieving well-balanced system-level optimization.

In summary, supported by both the detailed metric analysis and the aggregated results in Table II, the experimental results across three real-world datasets provide clear answers to the three research questions:

- RQ1: EF-DH achieves superior overall system performance compared with all single-stage and single-algorithm baseline methods.
- RQ2: EF-DH realizes an effective balance between

UAV exploration and other objectives across different task scales.

- RQ3: The dual-stage cooperation mechanism significantly improves worker-side temporal efficiency and value acquisition.

VI. Conclusion

This paper proposed the Co-UPP framework to address critical inefficiencies in MCS, which are primarily caused by a lack of real-time environmental awareness. We introduced a novel EF-DH algorithm by elevating the UAV from a simple data collector to a strategic intelligence asset. By evolutionarily fusing the global exploration capability of MOPSO with the robust convergence of NSGA-II, EF-DH effectively solves a dual-stage optimization problem. It first plans the UAV path to co-optimize task rewards and environmental value, and subsequently uses that intelligence to inform and optimize ground worker task assignment. Based on three real-world datasets, we conducted extensive comparative experiments to evaluate the proposed framework. The results confirm that EF-DH consistently outperforms all baseline algorithms in terms of key solution quality metrics. Most critically, it drastically reduced the Worker Maximum Time, achieving improvements of 29%–44% over other meta-heuristics and 513%–800% over the Greedy algorithm. Furthermore, it achieved the highest Worker Total Value and UAV Reward across all scenarios. Although it consumes more computation time than the simple Greedy method, EF-DH runs faster than NSGA-II-DH and EHTA (SA). These results verify that the EF-DH-enabled Co-UPP framework offers an effective and robust approach for optimizing practical complex MCS operations.

In future works, we will focus on expanding this framework to incorporate multimodal mobility datasets and real-time streaming dynamics.

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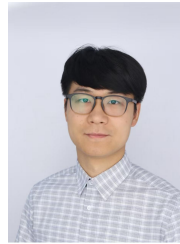


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